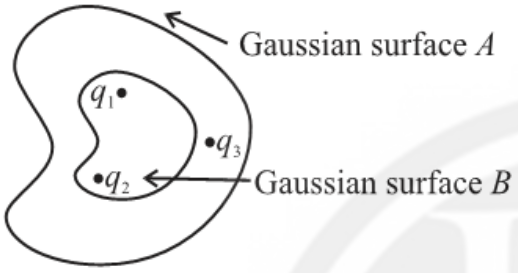
PARISHRAM 2.0 2026 (Class 12th)

PHYSICS

DPP: 08

Electric Charges and Fields

Q1 The electric flux for Gaussian surface *A* that enclose the charged particles in free space is; (Given q_1 = –14 *nC*, q_2 = 78.85 *nC*, q_3 = –56 *nC*)



(A) $10^3 Nm^2C^{-1}$

(B) $10^3 CN^{-1}m^{-2}$

(C) $6.32 \times 10^3 Nm^2C^{-1}$

(D) zero

Q2 Assertion (A): The no. of lines of force emanating from $1\mu C$ charge in vacuum is 1.13×10^5.

Reason (R): This follow from Gauss's theorem in electrostatics.

(A) Assertion (A) is True, Reason (R) is True; Reason (R) is a correct explanation for Assertion (A).

(B) Assertion (A) is True, Reason (R) is True; Reason (R) is not a correct explanation for Assertion (A).

(C) Assertion (A) is True, Reason (R) is False.

(D) Assertion (A) is False, Reason (R) is True.

Q3 A charge q is located at the center of a cube. The electric flux through any two consecutive faces will be

(A) $\frac{\pi q}{6(4\pi\varepsilon_0)}$

(B) $\frac{q}{6(4\pi\varepsilon_0)}$

(C) $\frac{2\pi q}{6(4\pi\varepsilon_0)}$

(D) $\frac{4\pi q}{3(4\pi\varepsilon_0)}$

Q4 Assertion (A): A charge q is placed in a cube of side b. The flux associated with the cube is independent of side length.

Reason (R): Gauss's law is independent of the size of the Gaussian surface.

(A) Assertion (A) is True, Reason (R) is True; Reason (R) is a correct explanation for Assertion (A).

(B) Both Assertion and Reason are incorrect

(C) Assertion (A) is True, Reason (R) is True; Reason (R) is not a correct explanation for Assertion (A).

(D) Assertion (A) is False, Reason (R) is True.

Q5 If $\oint \vec{E} \cdot \overrightarrow{dS} = 0$ over a surface, then:

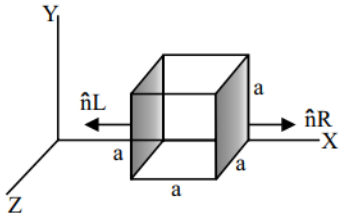
(A) The electric field inside the surface is necessarily uniform.

(B) The number of flux lines entering the surface must be equal to the number of flux lines leaving it.

(C) The magnitude of electric field on surface is constant.

(D) All the charges must necessarily be inside the surface.

Q6 The electric field components in the given figures are $E_x = \alpha x^{1/2}, E_y = E_z = 0$ in which $\alpha = 800 \mathrm{NC}^{-1}\ \mathrm{m}^{-1/2}$. The charge within the cube is [if net flux through the cube is $1.05 \mathrm{Nm}^2\mathrm{C}^{-1}$ (assume $\mathrm{a} = 0.1\ \mathrm{m}$)]

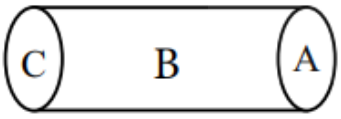


(A) $9.27 \times 10^{-12} \mathrm{C}$
(B) $11.25 \times 10^{12} \mathrm{C}$
(C) $6.55 \times 10^{-12} \mathrm{C}$
(D) $7.97 \times 10^{12} \mathrm{C}$

Q7 A uniformly charged and infinitely long line having a linear charge density ' λ ' is placed at a normal distance y from a point O. Consider a sphere of radius R with O as centre and $\mathrm{R} > \mathrm{y}$. Electric flux through the surface of the sphere is

(A) Zero
(B) $\frac{2\lambda R}{\varepsilon_0}$
(C) $\frac{2\lambda\sqrt{\mathrm{R}^2-\mathrm{y}^2}}{\varepsilon_0}$
(D) $\frac{\lambda\sqrt{\mathrm{R}^2+\mathrm{y}^2}}{\varepsilon_0}$

Q8 A hollow cylinder has a charge q within it. If ϕ is the electric flux in unit of volt meter associated with the curved surface B, the flux linked with the plane surface A in unit of volt meter will be



(A) $\frac{1}{2}\left(\frac{q}{\varepsilon_0}-\phi\right)$
(B) $2\varepsilon_0 q$
(C) $\frac{\phi}{3}$
(D) $\frac{q}{\varepsilon_0}-\phi$

Q9 The electric field in a certain region is acting radially outward and is given by $\mathrm{E} = \mathrm{Ar}$. A charge contained in a sphere of radius ' a ' centered at the origin of the field, will be given by

(A) $\mathrm{A}\varepsilon_0 \mathrm{a}^2$
(B) $4\pi\varepsilon_0 \mathrm{Aa}^3$
(C) $\varepsilon_0 \mathrm{Aa}^3$
(D) $4\pi\varepsilon_0 \mathrm{Aa}^2$

Q10 Long string with a charge of λ per unit length passes through an imaginary cube of edge a. The maximum flux of the electric field through the cube will be

(A) $\lambda a / \in_0$
(B) $\sqrt{2}\lambda a / \in_0$
(C) $6\lambda a^2 / \in_0$
(D) $\sqrt{3}\lambda a / \in_0$

Answer Key

Q1	A
Q2	A
Q3	D
Q4	A
Q5	B
Q6	A
Q7	C
Q8	A
Q9	B
Q10	D

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Net charge inside Gaussian surface A:

$$\begin{aligned} Q &= q_1 + q_2 + q_3 \\ &= (-14 + 78.85 - 56)\ \text{nC} = 8.85\ \text{nC} \\ &= 8.85 \times 10^{-9}\ \text{C}. \end{aligned}$$

Flux (Gauss's law):

$$\Phi = \frac{Q}{\varepsilon_0} = \frac{8.85\times10^{-9}}{8.85\times10^{-12}} = 10^3\ \text{N-m}^2/\text{C}.$$

Video Solution:

Q2 Text Solution:

Using Gauss's law, total flux (and hence the conventional "number of field lines") emerging from a point charge is

$$\Phi = \frac{Q}{\varepsilon_0}.$$

For $Q = 1\ \mu\text{C} = 1 \times 10^{-6}\ \text{C}$:

$$\Phi = \frac{1 \times 10^{-6}}{8.85 \times 10^{-12}} \approx 1.13 \times 10^5.$$

So the asserted value is correct, and it indeed follows directly from Gauss's theorem.

Option – both statements are true, and the reason correctly explains the assertion.

Video Solution:

Q3 Text Solution:

Flux through the whole cube (Gauss's law):

$$\Phi_{\text{total}} = \frac{q}{\varepsilon_0}$$

The cube has six identical faces, so flux per face:

$$\Phi_{\text{face}} = \frac{q}{6\varepsilon_0}$$

For any two consecutive (adjacent) faces:

$$\Phi_2 = 2\Phi_{\text{face}} = \frac{q}{3\varepsilon_0}$$

Video Solution:

Q4 Text Solution:

Gauss's law: $\Phi = \frac{q_{\text{enc}}}{\varepsilon_0}$ — flux depends only on the enclosed charge, not on the geometry or size of the closed (Gaussian) surface.

- **Assertion:** True — changing the cube's side b doesn't alter Φ.
- **Reason:** True — Gauss's law itself states the size/shape independence, thus explaining the assertion.

Video Solution:

Q5 Text Solution:

Flux $\oint \mathbf{E} \cdot d\mathbf{S} = 0$ means equal numbers of field lines enter and leave the closed surface, i.e. no net electric flux.

- It **does not** demand uniform $\mathbf{E}$, constant magnitude, or that charges reside inside; it only requires the balance of lines.

Option – the number of flux lines entering equals the number leaving.

Video Solution:



Q6 Text Solution:

Gauss's law $Q_{\text{enc}} = \varepsilon_0 \, \Phi_{\text{net}}$

$\Phi_{\text{net}} = 1.05 \text{ N-m}^2 \text{ /C},$

$\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{ /N-m}^2$

$$Q = (8.85 \times 10^{-12})(1.05) \text{ C} \approx 9.27 \times 10^{-12} \text{ C}$$

Video Solution:



Q7 Text Solution:

Inside the sphere the infinite line contributes only over the chord it cuts:

- Length of line inside sphere

  $L = 2\sqrt{R^2 - y^2}$
- Enclosed charge

  $Q_{\text{enc}} = \lambda L = 2\lambda\sqrt{R^2 - y^2}$

Flux (Gauss's law):

$$\Phi = \frac{Q_{\text{enc}}}{\varepsilon_0} = \boxed{\frac{2\lambda\sqrt{R^2 - y^2}}{\varepsilon_0}}$$

Video Solution:



Q8 Text Solution:

$$\Phi_A + \Phi_B + \Phi_C = \frac{q}{\varepsilon_0}$$

The charge is on the axis, so the two end caps are equivalent:

$\Phi_A = \Phi_C$.

Given $\Phi_B = \phi$:

$$2\Phi_A + \phi = \frac{q}{\varepsilon_0} \implies \boxed{\Phi_A = \tfrac{1}{2}\left(\frac{q}{\varepsilon_0} - \phi\right)}$$

Video Solution:



Q9 Text Solution:

Gauss's law: $q = \varepsilon_0 \, \Phi$

For a sphere of radius a

$$E = Ar \implies E(a) = Aa$$

Flux through the spherical surface

$$\Phi = E(a)\,(4\pi a^2) = Aa\,(4\pi a^2) = 4\pi A a^3$$

Enclosed charge

$$q = \varepsilon_0 (4\pi A a^3) = 4\pi\varepsilon_0 A a^3$$

Video Solution:

Q10 Text Solution:

$$\phi_E = \frac{\lambda(\text{Lenclosed})}{\epsilon_0}$$

Max. enclosed length = $\sqrt{3}a$

Video Solution:

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